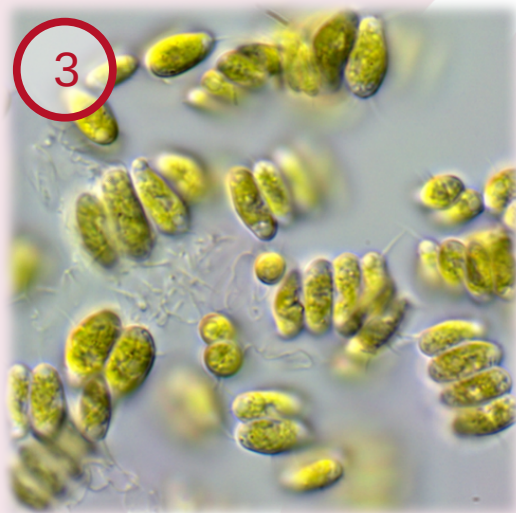
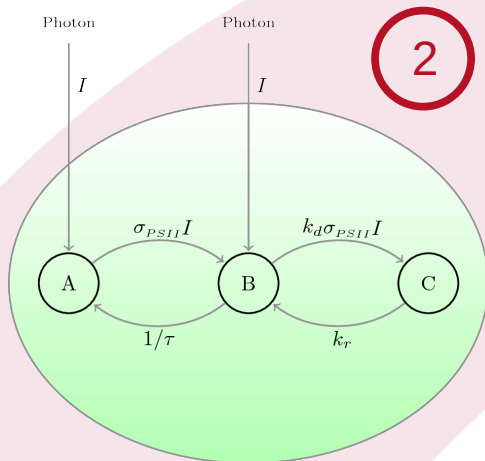


Fulbright France Call 2025

The research group:

The **Chair of Biotechnology of CentraleSupélec** is located in the European Center for Biotechnology and Bioeconomy (**CEBB**) (one hour from Paris). It leads research on three topics: biomaterials, **biotransformation**, and downstream processing.

In addition, the Chair has a strong expertise in **numerical modeling** applied to bioprocesses with a clear aim at **digital twinning** and **process upscaling**.



The proposed topics:

1. Microalgae biofaçade. This building/biotech' **hybrid technology** shows great promise. We evaluated its productivity using a **numerical model** (combining weather, realistic growth performances, ...). Exploring this topic, you would carry out a **Life Cycle Analysis** of the system.

2. New microalgal photosynthesis model. Accurately modeling microalgae photosynthesis is an important step towards intensified photobioreactor design. Unfortunately, the community is stuck with **models dating from the early 00s'**. This topic will **evaluate the downfalls** of these models and **propose new ones**, inspired by those of the higher plant community, for example.

3. Oxygen and pigment synthesis. Recent results suggest that oxygen level can substantially influence **fermentation-born** microalgae pigment content. This could be your pick if you are more into an **experimental topic**.

Open topic:

Do you have something in mind for a research stay in France, dealing with numerics or experiments, microalgae, fermentation, or something else? Please feel free to reach out for a discussion.

Contact

Dr. HDR. Victor Pozzobon – Head of the modeling team

✉: Victor.pozzobon@centralesupelec.fr

☎: +33 (0)3 52 62 05 08